Quick Start

Run Case Vault on one Windows computer and open its printed link on other devices in the allowed Wi-Fi subnet. MongoDB Atlas remains the database and needs internet access.

1. First-time setup

Install Python 3.12 and Node.js. In PowerShell, run the commands below. Check that the project's .env contains working Atlas settings; never share the file.

```
cd "C:\Users\James Berto\Downloads\case-vault_backend"
python -m venv venv
.\venv\Scripts\python.exe -m pip install -r requirements.txt
cd .\case-vault-frontend
npm.cmd ci
cd ..
```

If venv and node_modules already work, skip the install commands.

2. Start the system each time

Connect	Connect the host computer and client devices to the intended Wi-Fi. Keep the host on and online.
Run	Open PowerShell in the project folder and enter the LAN launcher command:

```
.\scripts\Start-CaseVaultLAN.ps1
```

Open	Wait for 'Application startup complete'. Copy the printed Case Vault LAN address into each client browser. The address can change after reconnecting to Wi-Fi.
Stop	Keep the hosting PowerShell window open. Press Ctrl+C there to stop the server.

3. Common startup problems

Script blocked	Run Set-ExecutionPolicy -Scope Process RemoteSigned in that PowerShell window, then retry.
Wi-Fi IP error	Run ipconfig and pass your current address and prefix, for example -LanIp 172.20.10.6 -PrefixLength 28 for mask 255.255.255.240.
Server error	Read the earlier error lines. Port 8000 may already be used by another running Case Vault server.
No client access	Check the printed IP, Windows Firewall, and Wi-Fi client isolation. See LAN.md for firewall details.

Navigation and Permissions

Sign in with your assigned account. The Case Vault logo opens the overview. Tabs vary by role; the key icon opens your own password settings, and Log Out ends your session.

Where to go

Case Library	Browse active cases, see counts, search by case number, type, creator, or tag, filter by status, and open a case. New Case appears if you can create cases.
Case Detail	Read case information, evidence, hashes, integrity results, version history, and assigned tags. Actions shown depend on your permissions.
Secure Submission	Upload a new PDF evidence record into an existing case. Create the case first if none is available.
Status	See your account and assigned permissions. Administrators see their dashboard on the Case Vault overview page.
Admin dashboard	Review the permission matrix, account list, online status, latest integrity attempt, and live activity ledger.

Built-in role guide

Administrator	Can create and view cases, upload/view evidence, change case status, manage tags and accounts, see audit activity, and archive cases or evidence.
Analyst	Can create/view cases, upload/view evidence, create/assign/remove tags, and verify/download evidence. Cannot change case status or archive records.
Reviewer	Can view cases and evidence, change case status, assign/remove existing tags, and verify/download evidence. Cannot create cases, upload files, or archive records.

Everyone can change their own password. Backend permissions control access even if a button is hidden or someone edits a URL. Custom roles depend on their database grants; the dashboard shows the current matrix.

Sessions

Each account has one active login. If it is open elsewhere, the login page shows an estimated countdown until the three-minute lease expires. Activity on the first device renews the lease; signing out there frees it sooner. The timer never signs in automatically. Log out when finished.

Cases, Evidence, and Tags

Create and find a case

In Case Library, select New Case. The number field previews a date-based case number. Enter a case type and create it. The backend assigns the final unique number. Open the case to review details or add evidence. The library excludes soft-deleted cases.

Submit evidence

Open Secure Submission or select Add Evidence from a case detail page. Choose an existing Associated Case, enter an evidence title and classification, then choose a PDF. Source device, description, and notes are optional. Select Upload Evidence and wait for confirmation.

The backend parses the PDF content and rejects malformed files; changing a filename to .pdf is insufficient. The maximum is 10 MiB. Upload stores the bytes, a SHA-256 baseline hash, a version record, and an audit event. It does not mean a later integrity check has happened.

Evidence versions and integrity

Download	Case Detail offers the current file and older versions through Version history. Replacement PDF uploads create a new version and preserve older files and hashes.
Verify	Select Verify on the current or an older version. The backend reads the stored file and compares its SHA-256 with that version’s upload baseline. This check runs only when requested.
Not checked	A baseline exists, but no completed comparison is recorded for that version.
Intact	The stored bytes matched the baseline at the displayed check time.
Mismatch detected	The stored bytes differed. The version is tamper flagged; the prior flag is not automatically cleared.
Check failed	The file could not be read or the comparison could not finish; this is an error, not proof of tampering.

Case status and tags

Reviewers and administrators can set a case to open, under review, or closed. These workflow values are separate from evidence integrity results.

Administrators and analysts can create reusable tag names. All built-in roles can view, assign, and remove tags on active cases. Removing an assignment keeps the tag available for other cases. Administrators can delete a tag globally, which removes it from all case associations.

Administration and Activity

Manage accounts

On the administrator overview, Add User creates an account with name, email, temporary password, and role. New emails must end in exactly @casevault.org. Select the eye icon on a user row to inspect or change that account’s role or active state. Deactivation prevents sign-in; reactivation permits it again. There is no account-delete action.

All users can select the key icon to change their own password. The current password and a new 8-128 character password are required. Common new passwords are rejected. Administrators may set a common temporary password during account creation; the user should replace it promptly.

Activity ledger

The admin dashboard shows recent recorded actions and receives live activity updates while connected. View All reveals search and filters. Entries identify the actor, action, time, and related case or evidence when recorded. Passwords, tokens, and PDF contents are not displayed in the ledger.

Archive (soft delete)

Administrators can archive a case or evidence from Case Detail. Normal lists and evidence operations then hide the archived record. Metadata and stored files remain retained; admin-only API endpoints can inspect deleted metadata. Archiving is recorded in the ledger. It is not a permanent file purge.

Errors and good habits

403	Your role is not allowed to perform that action. Ask an administrator to review the permission matrix.
404	The item may be missing or archived; normal pages do not expose soft-deleted records.
409	A case number, tag, or assignment may already exist, or that account is signed in elsewhere.
400 / 413	The PDF content or file type is invalid, or the file exceeds the 10 MiB limit.
422	Check required fields, text lengths, and @casevault.org email for new accounts.
Connection	Confirm the LAN server still runs, the host is online, and MongoDB Atlas is reachable.

For complete setup and firewall instructions, read LAN.md. For technical API and migration details, read BACKEND.md. This guide describes the current interface and built-in roles.

Development Tools: Backend

What each server-side tool does, why it is used, and how it fits into a request.

Python 3.12

Purpose: Language of the backend and maintenance scripts.
How: Runs app code, receives upload bytes, calls MongoDB, and creates API responses.

FastAPI

Purpose: Framework that defines the backend's HTTP endpoints and permissions.
How: Maps a path such as /cases to a Python function, validates inputs, and returns JSON or a file.

Uvicorn

Purpose: Server process that makes FastAPI reachable in a browser.
How: Listens on a host IP and port, passes each request to the app, and sends back its response.

Pydantic + pydantic-settings

Purpose: Checks input and output shapes and reads configuration.
How: Rejects invalid fields before business logic runs; loads MONGO_URI, DB_NAME, and JWT_SECRET from .env.

Beanie + Motor + PyMongo

Purpose: Document models and async MongoDB access.
How: Beanie maps Python models to collections; Motor/PyMongo execute queries, indexes, and transactions.

MongoDB Atlas

Purpose: Hosted database for accounts, cases, versions, sessions, tags, and audit entries.
How: The server connects over the internet; metadata transactions keep related writes consistent.

GridFS

Purpose: MongoDB file storage for evidence PDFs.
How: Stores file bytes in chunks, identified by a file ID kept in each evidence version.

pypdf + hashlib SHA-256

Purpose: PDF structure check and evidence integrity baseline.
How: pypdf parses uploaded pages; hashlib hashes the received bytes and rehashes stored bytes on Verify.

passlib/bcrypt + python-jose

Purpose: Password protection and signed login tokens.
How: Passwords are checked against stored hashes; a signed JWT identifies the logged-in user.

Request path: browser -> Uvicorn/FastAPI -> validation and permissions -> MongoDB/GridFS -> response.

Development Tools: Frontend

Tools that build the interface and help developers run, check, and track changes.

Node.js + npm

Purpose: Runs JavaScript build tools and manages frontend packages.
How: npm.cmd ci installs exactly the packages in the lockfile; npm scripts start checks and builds.

React

Purpose: Renders pages, forms, tables, and live status in the browser.
How: Components hold page state and redraw when API results or user input changes.

React Router

Purpose: Controls movement between Case Vault screens.
How: Routes select the page; permission guards block pages an account cannot use.

Vite

Purpose: Frontend development server and production build tool.
How: npm.cmd run dev serves editable code locally; npm.cmd run build creates files for LAN mode.

CSS + installed Tailwind tooling

Purpose: Styles the interface at different screen sizes.
How: Page CSS sets colors and layouts; Tailwind is installed in the frontend build configuration.

Lucide React

Purpose: Reusable icons in buttons, tabs, and status panels.
How: Components draw vector icons such as the key, folder, and search symbols.

PowerShell LAN launcher

Purpose: Starts the frontend and backend together for Wi-Fi access.
How: Builds Vite, detects or accepts the Wi-Fi IP, sets the allowed subnet, then starts Uvicorn.

pytest + httpx

Purpose: Backend automated checks and test HTTP requests.
How: pytest runs test cases; httpx sends requests to the API in isolated tests.

ESLint

Purpose: Frontend source checker.
How: npm.cmd run lint finds JavaScript/React mistakes before use or deployment.

Git

Purpose: Source history and branch management.
How: Records code changes so developers can review or restore earlier versions; it is not the database.

Key files: requirements.txt lists Python packages; package.json lists frontend packages and commands; .env stores private local settings. Do not share .env.

HTTP Responses

The number in an API response tells you whether an action succeeded and why it failed.

Success: 2xx

200 **OK**
The request worked and returned data, for example login, case listing, or a downloaded PDF.

201 **Created**
A new record was saved, for example an account, case, evidence item, version, or tag.

204 **No Content**
The action worked but returns no body, for example logout, password change, or archive.

Request or access problems: 4xx

400 **Bad Request**
The uploaded file is not a valid PDF, or the current password is wrong.

401 **Unauthorized**
Credentials are wrong or the login token/session has ended. Sign in again.

403 **Forbidden**
Your role cannot perform the action, the account is deactivated, or LAN access is denied.

404 **Not Found**
The case, evidence, version, user, or tag is absent or hidden by soft deletion.

405 **Method Not Allowed**
The URL exists but the request used the wrong method, such as GET instead of POST.

409 **Conflict**
A duplicate case/tag/email/assignment exists, or the account is signed in elsewhere.

413 **Content Too Large**
The uploaded PDF exceeds the 10 MiB evidence-file limit.

422 **Validation Error**
A required field, length, date range, role, or new-account email domain is invalid.

Server or storage problems: 5xx

500 **Internal Server Error**
The server cannot complete the request, for example a missing permission configuration.

503 **Service Unavailable**
Evidence verification could not read the stored file; no comparison was completed.

No HTTP code at all usually means the browser could not reach the server. Check the LAN link, host, Wi-Fi, firewall, and Atlas connection. The response's detail text gives the specific cause when available.